Calibrating Geometric Consistency for Reliable 3D Scene Graph Relations

Anonymous submission

Abstract

3D Scene Graphs represent objects and relations in a form useful for spatial reasoning, but relation predictions can remain unreliable even when predicted predicates appear semantically plausible. We study a relation-level failure mode: semantic relation confidence is not necessarily calibrated to the 3D geometry of the same object pair, so a predictor can rank physically inconsistent relations highly. To expose and reduce this failure, we introduce a calibrated geometry-consistency evaluation and re-ranking framework for geometry-checkable relation families. The framework standardizes prediction rows across relation sources, preserves subject-object identity, joins explicit 3D evidence, estimates $p^{\text{geom\_valid}}$, and reports family-conditional calibrated geometry risk under a recall-violation protocol. The main GeoCalib score ranks by semantic confidence multiplied by a relation-family-conditioned geometry-validity probability; pooled calibration and hard-rule variants are retained as ablations and diagnostics. On the full official validation split, GeoCalib re-ranking improves VL-SAT R@50/R@100 from 0.9272/0.9635 to 0.9288/0.9683 while reducing Violation@100 from 0.0476 to 0.0333. On the main open-vocabulary relation-source case study, the Open3DSG full-validation recovery branch, the same family-conditional score improves R@100 from 0.5161 to 0.6047 while reducing Violation@100 from 0.1242 to 0.0341 under explicit selected-checkpoint and recovery-policy caveats. GT positive/counterfactual checks separate valid and invalid geometry with AUROC/AUPRC of 0.9772/0.9729, and controls rule out geometry-only, distance-only, shuffled-geometry, and wrong-pair explanations. These results support a scoped relation-reliability claim for measured geometry-checkable 3D Scene Graph relations, not a broad open-vocabulary graph generation claim.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be reliable in the physical scene, not only plausible at the category or language level. A graph edge such as `standing on`, `next to`, or `higher than` is useful only if it describes the actual subject-object pair in 3D.

We study a failure mode in which semantic relation predictors rank plausible relation labels that are geometrically inconsistent for the actual object pair. A relation can sound plausible for two object categories while being contradicted by their contact evidence, distance, or vertical arrangement in the reconstructed scene. This mismatch matters more as 3D Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: semantic relation confidence is not necessarily a calibrated estimate of relation-level physical consistency. A high relation score can behave as a semantic plausibility score without indicating whether the same subject-object pair satisfies the relation in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a confident symbolic relation that contradicts the underlying world state.

This motivates a calibrated geometry-consistency evaluation and re-ranking framework. The framework treats an existing predictor as a relation source, standardizes its outputs into identity-preserving prediction rows, joins object-pair geometry evidence by subject and object instance identifiers, estimates calibrated geometry-validity risk, and evaluates relation rankings with both exact-label recall and geometric violation. The main GeoCalib score is family-conditional: semantic confidence is penalized by a relation-family-specific estimate of geometry validity. This design is necessary because hard physical rules, pooled calibration, and family-conditioned calibration answer different questions: rule-verified variants diagnose strict consistency, pooled calibration is an ablation, and family-conditioned calibration is the paper's primary recall-violation score. Figure 1 summarizes this path.

We evaluate the framework on geometry-checkable relation families: `support_contact`, `proximity`, and `relative_vertical`. These families are selected because their validity can be tested from reconstructed object-

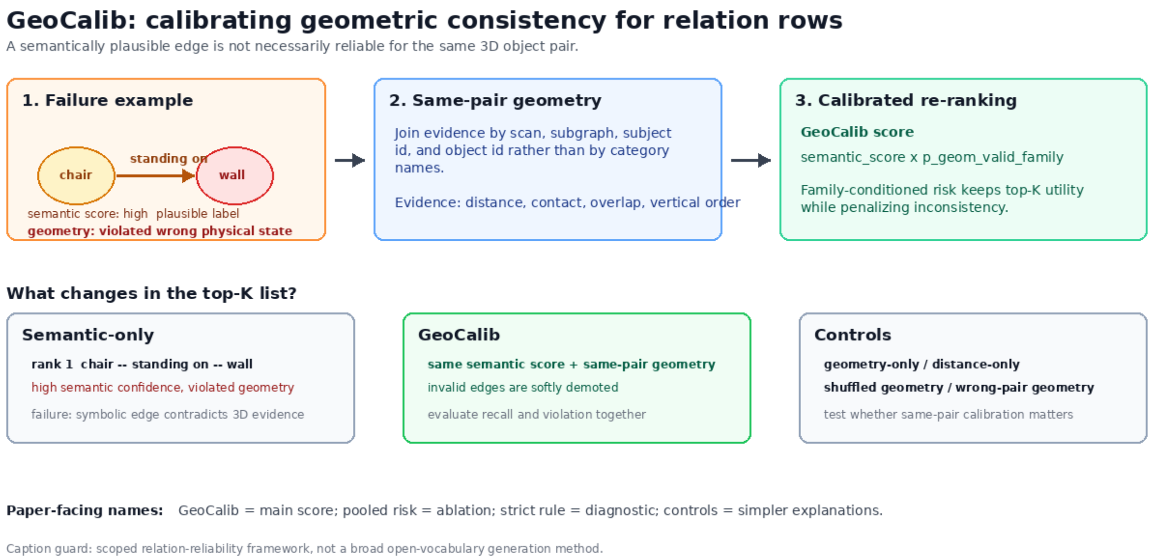

Figure 1: **GeoCalib in one relation-row example.** A semantically plausible relation can be physically inconsistent for the same object pair. GeoCalib preserves row identity, joins pair geometry, estimates family-conditioned geometry validity, and re-ranks top-K relations while reporting recall and violation together.

pair geometry, they cover contact, metric proximity, and vertical ordering, and they appear in both relation sources with a fixed exact-label denominator. Open3DSG is the main open-vocabulary relation source case study, and VL-SAT is a controlled reproduced closed-set anchor within the same geometry-checkable family scope (Wang et al. 2023; Koch et al. 2024b). Controls test whether the effect can be explained by geometry-only ranking, a generic distance heuristic, shuffled geometry, or wrong-pair geometry. GT-positive and counterfactual checks test the geometry-validity signal independently from prediction re-ranking, while qualitative failure analysis shows how semantic plausibility diverges from physical consistency.

Our contributions are threefold:

1. We identify and formalize a relation-level reliability failure in 3D Scene Graph prediction: semantic relation confidence can rank plausible predicates that are physically inconsistent because it is not calibrated to object-pair geometry.
2. We introduce a calibrated geometry-consistency evaluation and re-ranking framework that standardizes prediction rows, joins identity-preserving 3D evidence, estimates calibrated geometry validity, and re-ranks by a family-conditional geometry-risk score while retaining pooled and rule-verified variants as ablations and diagnostics.
3. We define a recall-violation evaluation protocol for geometry-checkable relation families, including exact-label R@K, Violation@K, GT-positive/counterfactual verifier evaluation, and geometry identity controls.

We validate these contributions with Docker-reproducible evidence on VL-SAT and Open3DSG, including nontriviality controls, Open3DSG failure analysis, and explicit reproduction caveats. The claim is intentionally scoped: calibrated geometry-consistency evaluation and re-ranking measures and improves relation reliability for geometry-checkable 3DSSG families. It does not claim general open-vocabulary 3D Scene Graph generation improvement, arbitrary-source generality, or guaranteed physical correctness.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs represent indoor scenes as object-relation structures for spatial reasoning. The 3RScan/3DSSG setting established predicate recall for 3D relation prediction (Wald et al. 2020), and later methods use edge reasoning, point-cloud features, multimodal cues, and spatial knowledge to improve closed-set predicates (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a). These methods already use geometry; calibrated geometry-consistency evaluation and re-ranking instead targets the post-source reliability gap that a high semantic relation score need not be calibrated to physical validity for the same object pair.

Open-vocabulary 3D Scene Graphs and VLM-based scene understanding. Recent work extends 3D scene graphs toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, plausible text labels must still refer to object pairs that satisfy the corresponding spatial relation.

ZING-3D, VIZOR, and Open-World 3DSG-RAG further connect VLMs, zero-shot graph construction, and retrieval-augmented reasoning (Saxena and Chiun 2025; Madhavaram et al. 2026; Yu et al. 2025). We cite them as trend and boundary evidence, not direct calibrated geometry-consistency evaluation and re-ranking baselines. Qwen-VL remains an optional semantic-source extension unless promoted with the same Docker, denominator, metric, and audit treatment.

Geometry-aware relation reasoning and semantic-geometric fusion. A broad set of 3D reasoning methods already uses geometry: edge features, support/contact patterns, affordance geometry, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Huang et al. 2025; Shao et al. 2025; Xie, Pagani, and Stricker 2024). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness uses visual-geometric witnesses, calibrated witness quality, and witness-consistent decoding for open-vocabulary 3D scene graph generation (Nguyen et al. 2026). calibrated geometry-consistency evaluation and re-ranking therefore does not claim novelty from visual-geometric evidence or calibration alone. It studies a different post-source contract: identity-preserving geometry joins, recall/violation operating points, and controls over existing relation-source outputs.

Reliability and calibration for structured relation outputs. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is physically valid in the reconstructed scene. calibrated geometry-consistency evaluation and re-ranking instantiates this reliability view for geometry-checkable 3DSSG families by standardizing prediction rows, joining same-pair 3D evidence, estimating $p^{\text{geom\_valid}}$, and reporting recall with violation.

Problem Formulation

Let a relation source produce a set of candidate relation rows for a 3D scene. Each row is written as

$$r_i = (\text{scan}_i, \text{subgraph}_i, s_i, o_i, p_i, a_i, \text{score}_i^{sem}), \tag{1}$$

where s_i and o_i are subject and object instance identifiers, p_i is the predicate label, a_i is the mapped predicate family, and score_i^{sem} is the source semantic confidence. The tuple $(\text{scan}_i, \text{subgraph}_i, s_i, o_i)$ is the identity key used to preserve the object pair across prediction, geometry, and ground-truth joins.

For each prediction row, we join object-pair geometry evidence

$$g_i = G(\text{scan}_i, \text{subgraph}_i, s_i, o_i). \tag{2}$$

The evidence can include OBB features, point/local contact evidence, distance, vertical ordering, support/contact subtype information, and missing-evidence reason codes. calibrated geometry-consistency evaluation and re-ranking evaluates only geometry-checkable relation families:

$$\mathcal{A} = \{\texttt{support\_contact}, \texttt{proximity}, \texttt{relative\_vertical}\}. \tag{3}$$

Rows outside these families are preserved in exports but excluded from the measured violation claim.

The geometry verifier assigns a row-level status $v_i \in \{\text{satisfied}, \text{uncertain}, \text{violated}\}$, and the calibrator estimates pooled and family-conditioned geometry-validity probabilities:

$$p_i^{\text{geom\_valid}} = P(v_i \neq \text{violated} \mid g_i, a_i). \tag{4}$$

The main GeoCalib score uses the family-conditioned version of this probability, denoted $p^{\text{geom\_valid,family}}$, while the pooled $p^{\text{geom\_valid}}$ score remains a calibrated ablation. These scores are reliability signals, not proofs of correctness. For this reason, the paper reports the main family-conditional score together with pooled, rule-verified, and control variants.

The main recall metric is exact predicate-label R@K over the fixed in-scope ground-truth denominator. Family grouping defines the geometry-checkable reliability scope and violation analysis; it does not relax predicate-label matching. Violation@K is the fraction of top-K in-scope predicted rows whose joined geometry is judged violated. Top-K is formed per evaluated subgraph/source ranking after the measured-family filter; satisfied, uncertain, and violated rows form the violation denominator, while unsupported or missing-join rows are excluded from that denominator and reported through coverage caveats. Only rows with verification status `violated` count as violations. We report recall retention and violation reduction together so that violation reduction cannot be interpreted without its recall tradeoff.

Method

calibrated geometry-consistency evaluation and re-ranking is a framework for 3D scene graph relation predictions. It is not a replacement relation predictor and not a standalone hand-coded verifier. The geometric checks define the measurable validity target for families with observable physical constraints; the method contribution is the identity-preserving row contract, calibrated $p^{\text{geom\_valid}}$ signal, operating-point analysis, and controls that test whether geometry is used as a relation-level reliability signal rather than as a generic filter. The framework is built around four design requirements: preserve relation-row identity, join explicit object-pair geometry, calibrate geometry validity, and evaluate recall and violation jointly. Figure 1 presents these requirements as a pipeline from relation-source outputs to calibrated operating points.

Relation-Row Standardization

Different relation sources expose predictions in different formats. We convert each source output into a shared row contract:

$$(\text{scan}, \text{subgraph}, s, o, p, a, \text{score}^{sem}, \text{source}). \tag{5}$$

This row is the unit of comparison across VL-SAT, Open3DSG, controls, and future optional sources. The standardized row keeps the semantic score from the source model but adds the fields needed for identity-preserving geometry joins and denominator-stable evaluation.

Identity-Preserving Geometry Join

Geometry is joined by $(\mathrm{scan}, \mathrm{subgraph}, s, o)$, not by category names or unordered object-pair statistics. This matters because the same predicate can be plausible for many object categories while being false for a particular instance pair. The wrong-pair and shuffled-geometry controls directly test this design choice by breaking or perturbing object-pair identity while preserving parts of the geometry distribution.

Family-Specific Geometry Checks

calibrated geometry-consistency evaluation and re-ranking uses relation-family-specific checks for `support_contact`, `proximity`, and `relative_vertical`. For support/contact, the evidence includes point/local contact, vertical support, OBB overlap, and support subtypes. For proximity, the evidence includes object distance and pair scale. For relative vertical relations, the evidence includes centroid and extent ordering along the vertical axis. The verifier returns violated only when the evidence is strong enough; otherwise ambiguous or weak geometry can be marked uncertain. We do not apply this protocol to predicates whose correctness depends primarily on function, social context, affordance, or language pragmatics, because those predicates do not have a stable geometric validity test from the available reconstruction alone.

Family-Conditional Calibrated Risk Re-Ranking

The main GeoCalib score uses a frozen relation-family-conditioned calibrator:

$$p_i^{\mathrm{geom\_valid,family}} = C_{a_i}(\phi(g_i)), \tag{6}$$

where $\phi(g_i)$ is the geometry feature vector and C_{a_i} is the calibrator for the mapped relation family. This is the paper's primary calibration design: support/contact, proximity, and vertical relations need not share one pooled geometry-risk surface. The pooled calibrator,

$$p_i^{\mathrm{geom\_valid}} = C_{\mathrm{pooled}}(\phi(g_i)), \tag{7}$$

is retained as the `pooled_risk` ablation. Both calibrators are trained on train-dev calibration positives and counterfactual negatives. The predicate-family map, hard-rule thresholds, counterfactual construction, and calibrator files are fixed from train-dev calibration artifacts before held-out source-result reporting; held-out prediction rows are not used to retune them.

We use geometry validity as a soft risk term. For the main score, $R_i^{geom} = -\log p_i^{\mathrm{geom\_valid,family}}$ and

$$U_\lambda(i) = \log \mathrm{score}_i^{sem} - \lambda R_i^{geom}. \tag{8}$$

The reported GeoCalib score fixes $\lambda = 1$, giving

$$\mathrm{score}_i^{\mathrm{GeoCalib}} = \mathrm{score}_i^{sem} \cdot p_i^{\mathrm{geom\_valid,family}}. \tag{9}$$

This preserves the source ranking signal while continuously penalizing calibrated inconsistency; λ is not tuned on held-out source results. We report `family_conditional_risk` as the main score, `probabilistic_recalibrated` as the pooled calibrated-risk ablation $\mathrm{score}_i^{sem} \cdot p_i^{\mathrm{geom\_valid}}$, and `rule_verified_point_subtype` as a hard zero-violation diagnostic.

Controls

The control conditions test whether the result can be explained by simpler signals. Geometry-only ranking uses $p^{\mathrm{geom\_valid}}$ alone and is therefore distinct from the pooled calibrated-risk ablation, which still multiplies semantic confidence by $p^{\mathrm{geom\_valid}}$. Distance-only ranking tests whether a generic spatial heuristic explains the result. Shuffled-geometry and wrong-pair-geometry controls test whether relation-level object-pair identity is necessary. If these controls perform worse than the calibrated setting, the result supports the framework design rather than a generic "add geometry" explanation.

Experimental Setup

This section fixes scope before results because calibrated geometry-consistency evaluation and re-ranking's claim depends on denominator transparency as much as metric values. Open3DSG is the main open-vocabulary relation-source case study, while VL-SAT is a controlled reproduced closed-set anchor. Families and denominators are fixed before source-result reporting, yielding a predeclared physically checkable slice: 157 scans, 548 contexts, 36,808 directed pairs, 957,008 VL-SAT prediction rows, 11,254 ground-truth rows, and 3,972 in-scope GT relations. Table 1 records the denominator and source outputs used by Table 2.

Open3DSG is reported as a Docker-reproduced source-output reliability case study, not an official leaderboard reproduction. The selected checkpoint is the official non-averaged BLIP route `epoch=13-step=13104.ckpt`, chosen by train-dev validation loss before held-out source-result inspection. Its runtime uses filtered train/dev subgraphs, and the main full-validation branch reaches 548/548 contexts with `OPEN3DSG_MIN_VISIBLE_OBJECTS=2` plus relaxed view regeneration for two scans. The 533/548 covered branch and historical old 377/388 vs R2 388/388 branch remain sensitivity evidence only.

Condition names are fixed before reporting. `family_conditional_risk` is the main GeoCalib score, ranking by $\mathrm{score}^{sem} \cdot p^{\mathrm{geom\_valid,family}}$. `probabilistic_recalibrated` is the pooled calibrated-risk ablation $\mathrm{score}^{sem} \cdot p^{\mathrm{geom\_valid}}$. Geometry-only controls rank by $p^{\mathrm{geom\_valid}}$ without semantic confidence and are reported separately from pooled calibrated re-ranking.

Source boundaries are enforced in prose rather than as a main result table. Open3DSG supports the open-vocabulary relation-source case study; VL-SAT supports the controlled-anchor comparison. FROSS remains runtime-blocked, Qwen-VL is an optional third-source extension,

Table 1: **Full official calibrated geometry-consistency evaluation and re-ranking evaluation scope.** Recall is exact predicate-label recall over the 3,972-row in-scope GT denominator; family grouping is used for reliability and violation reporting, not for relaxing predicate matches.

Item	Count	Role in claim
Validation scans	157	Full official validation scan set.
Subgraphs / contexts	548	Object-pair identity denominator.
Directed object pairs	36,808	Candidate relation-row universe.
Ground-truth rows	11,254	Full validation relation annotation rows.
In-scope GT rows	3,972	Exact-label denominator for the three measured relation families.
VL-SAT prediction rows	957,008	Controlled reproduced closed-set anchor output.
Open3DSG prediction rows	695,916	Main open-vocabulary source output.
Geometry-scored rows	160,596	Open3DSG geometry-checkable rows joined to calibrated evidence.

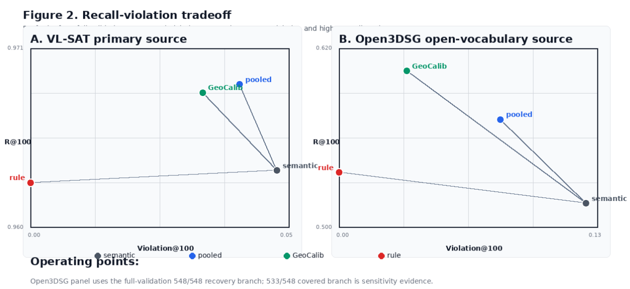

Figure 2: **Recall-violation tradeoff.** Lower Violation@100 and higher R@100 are better. Open3DSG is reported on the full-validation recovery branch; 533/548 and historical 377/388 vs R2 388/388 branches are sensitivity evidence only.

and broad open-vocabulary 3DSSG generation improvement is not claimed without the same Docker, denominator, metric, and audit treatment. All paper-body experiment results are generated through the Docker experiment root. Host-only outputs are not promoted to paper evidence.

Results and Discussion

Table 2 asks whether calibrated geometry consistency reduces violations without collapsing recall. On Open3DSG, the semantic source has R@50/R@100=0.4096/0.5161 and Violation@50/Violation@100=0.1386/0.1242. The lower-rank diagnostic is more severe: Violation@5/Violation@10=0.5131/0.3255, showing that semantically plausible but geometrically inconsistent relations can dominate the very top of the Open3DSG ranking. Pooled risk, the pooled calibrated ablation, lowers violations to 0.0606/0.0811 while moving recall to 0.3975/0.5723. The main GeoCalib score gives the strongest tradeoff: 0.4658/0.6047 recall with 0.0286/0.0341 violation. At low K, GeoCalib improves Open3DSG R@5/R@10 from 0.0368/0.1002 to 0.0984/0.1921 while reducing Violation@5/Violation@10 to 0.0420/0.0482.

VL-SAT provides the controlled anchor. The main GeoCalib score improves R@50/R@100 from 0.9272/0.9635 to 0.9288/0.9683 and reduces Violation@100 from 0.0476 to 0.0333. The pooled ablation reaches slightly higher R@50/R@100=0.9305/0.9688 but leaves a higher Violation@100=0.0404, so it is reported as a recall-favoring ablation rather than the main reliability score. VL-SAT already has low Violation@5/Violation@10; the main score still reduces those rates from 0.0029/0.0082 to 0.0011/0.0051. Strict rule verification is a zero-violation diagnostic, not the default method.

A Docker subgraph bootstrap with 1,000 resamples gives the Open3DSG `family_risk` operating point a R@100 delta of +8.86 percentage points (95% CI +6.69,+10.96) and a Violation@100 delta of -9.01 points (95% CI -9.49,-8.53) relative to `semantic_only`. VL-SAT deltas are smaller but still favorable: `family_risk` changes R@100 by +0.48 points (95% CI +0.11,+0.93) and reduces Violation@100 by -1.43 points (95% CI -1.60,-1.28). We use these intervals as evaluation-context uncertainty checks, not as repeated-training variance.

Table 3 rules out simpler alternatives. Geometry-only is not the same as pooled calibrated re-ranking: it removes semantic confidence entirely. Distance-only is weaker, and shuffled/wrong-pair controls show that object-pair identity is necessary. Thus calibrated geometry-consistency evaluation and re-ranking is not merely ranking by geometry, distance, or hard rules; calibrated same-pair evidence matters.

The GT-based verifier check tests the geometry signal independently: GT positives remain nonviolated at 0.9965, counterfactual negatives are nonsatisfied at 0.9673, and $p^{\text{geom\_valid}}$ separates 7,944 rows with AUROC/AUPRC 0.9772/0.9729 and Brier 0.0543. This distinguishes the calibrated reliability signal from a purely hand-written validity script.

Structured audit and visual sanity checks support failure interpretation but are not large-scale human evaluation: a 250-row structured audit reaches quality-issue precision 0.8933, and a 50-row visual spot-check has target-bucket quality-issue rate 0.9333 with contradiction rate 0.0333.

Together, the metrics and controls support a calibrated reliability-layer interpretation, not a new relation generator or broad open-vocabulary SOTA claim. Open3DSG comparisons share the same selected checkpoint, row contract, full-validation denominator, and recovered source payload. The historical R2 branch stays outside Table 2: it covers 388/388 contexts and changes old 377/388 R@100 by only

Table 2: **Main source results.** Rows use the full official 3,972-row three-family denominator. Open3DSG uses the selected non-averaged checkpoint on the 548/548 recovery branch; 533/548 and historical 377/388 vs R2 388/388 branches are sensitivity evidence only. We report the fixed low-K grid $K \in \{5, 10, 20, 50, 100\}$; $K = 1$ is excluded as a noisy sanity check. Residual calibration risk is reported in the failure analysis rather than hidden in the table. GeoCalib is the main family-conditional calibrated-risk score; pooled risk is the pooled `probabilistic_recalibrated` ablation.

		R@K					Violation@K				
Source	Condition	@5	@10	@20	@50	@100	@5	@10	@20	@50	@100
Open3DSG	Semantic	0.0368	0.1002	0.1991	0.4096	0.5161	0.5131	0.3255	0.2088	0.1386	0.1242
Open3DSG	Pooled risk	0.0826	0.1581	0.2603	0.3975	0.5723	0.0628	0.0699	0.0654	0.0606	0.0811
Open3DSG	Strict rule	0.0707	0.1314	0.2422	0.4295	0.5368	0.0000	0.0000	0.0000	0.0000	0.0000
Open3DSG	GeoCalib	0.0984	0.1921	0.3291	0.4658	0.6047	0.0420	0.0482	0.0441	0.0286	0.0341
VL-SAT	Semantic	0.4194	0.6322	0.8074	0.9272	0.9635	0.0029	0.0082	0.0142	0.0268	0.0476
VL-SAT	Pooled risk	0.4154	0.6322	0.8107	0.9305	0.9688	0.0015	0.0071	0.0120	0.0229	0.0404
VL-SAT	Strict rule	0.4197	0.6317	0.8074	0.9257	0.9627	0.0000	0.0000	0.0000	0.0000	0.0000
VL-SAT	GeoCalib	0.4162	0.6309	0.8087	0.9288	0.9683	0.0011	0.0051	0.0109	0.0206	0.0333

Table 3: **Controls and diagnostics.** These rows separate the main semantic-plus-geometry score from simpler explanations. Geometry-only ranks by $p^{\mathrm{geom\_valid}}$ without semantic confidence; pooled risk keeps semantic confidence and uses the pooled calibrator. Shuffled and wrong-pair controls break identity-preserving geometry evidence.

Source	Condition	R@100	Violation@100	Interpretation
Open3DSG	GeoCalib	0.6047	0.0341	Main family-conditioned semantic-plus-geometry score.
Open3DSG	Geometry-only	0.5116	0.0865	Geometry alone does not recover the main tradeoff.
Open3DSG	Distance-only	0.5038	0.1071	Generic distance is weaker than calibrated geometry risk.
Open3DSG	Shuffled geometry	0.2543	0.1998	Breaking row-level geometry destroys reliability.
Open3DSG	Wrong-pair geometry	0.2331	0.1985	Same-pair identity is necessary.
VL-SAT	GeoCalib	0.9683	0.0333	Main family-conditioned semantic-plus-geometry score.
VL-SAT	Geometry-only	0.5184	0.0711	Geometry alone is not a useful ranking source.
VL-SAT	Distance-only	0.5554	0.0981	Distance is a poor proxy for relation reliability.
VL-SAT	Shuffled geometry	0.9494	0.0588	Violations increase when geometry identity is broken.
VL-SAT	Wrong-pair geometry	0.9529	0.0601	Wrong object-pair evidence is worse than GeoCalib.

about +0.28 points, so the old missing-context caveat did not create the trend. Its clean-return raw files are row-equivalent after excluding run metadata, although the process-level teardown exit-137 caveat remains.

Failure analysis explains why the Open3DSG metrics change. The full-validation branch produces 82,155 diagnostic rows with zero validation errors; in the 36-case queue, 25 cases are demoted by geometry-aware re-ranking. Figure 3 shows distance, vertical-order, and support-plane contradictions that make symbolic edges unreliable for downstream reasoning.

The same analysis exposes residual risk: eight of 36 sampled rule-violated cases still have $p^{\mathrm{geom\_valid}} > 0.9$. Thus $p^{\mathrm{geom\_valid}}$ is not a hard validity label, and the paper reports the main family-conditional score alongside pooled and rule-verified variants.

Limitations

The study is scoped to geometry-checkable relation families. Functional, social, affordance, and language-pragmatic predicates are outside the calibrated geometry-consistency evaluation and re-ranking metric claim. The selected families cover contact/support, proximity, and vertical ordering, where reconstructed geometry supports falsifiable checks. Because evaluation uses GT-object relation rows and exact predicate-label recall, even Open3DSG is relation-reliability evidence rather than full open-vocabulary generation. We also audited broader horizontal relations and did not promote them: front/behind remains frame-ambiguous, and a left/right-only variant fails the train/dev strict gate because dev contradictions concentrate in repeated-object and orthogonal-axis-dominant cases. These failed expansion gates are treated as scope-boundary evidence rather than hidden negative results.

Open3DSG uses the selected official non-averaged checkpoint but is not a leaderboard reproduction. The paper-facing result uses a documented recovery branch to cover 548/548 contexts, while the 533/548 full-validation branch and historical 377/388 vs R2 388/388 comparison remain sensitivity evidence. These choices preserve the scoped within-source comparison but block broad Open3DSG SOTA or arbitrary-source generality claims.

The visual sanity check and qualitative case inspection are reviewer-defense and failure-mechanism evidence, not large-scale independent human audits. The quantitative claim remains tied to the fixed denominator and Docker-generated tables.

Finally, $p^{\mathrm{geom\_valid}}$ and $p^{\mathrm{geom\_valid,family}}$ are calibrated re-

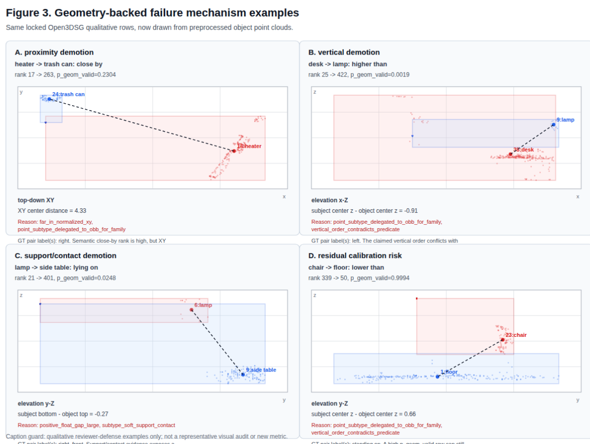

Figure 3: **Geometry-backed qualitative failures.** Open3DSG examples show proximity, relative-vertical, and support/contact cases where semantic plausibility conflicts with object-pair geometry. The rightmost case illustrates residual calibration risk.

liability scores, not proof of correctness. Residual high-confidence violations motivate attachment-style physical relations before broader functional predicates, plus larger visual audits, modern VLM sources, and downstream planning, navigation, or alignment tests.

Conclusion

This paper studies a scoped reliability failure in 3D Scene Graph relation prediction: semantic relation confidence can rank plausible predicates that are physically inconsistent for the same object pair. We address this failure with a calibrated geometry-consistency evaluation and re-ranking framework that standardizes relation rows, preserves subject-object identity, joins explicit 3D evidence, estimates $p^{\mathrm{geom\_valid}}$, and reports recall and geometric violation together.

On Open3DSG as the main open-vocabulary relation-source case study and VL-SAT as a controlled anchor, geometry-consistency reduces violations under measurable recall tradeoffs. GT/counterfactual checks, identity controls, and qualitative failures support the central claim: the contribution is not merely adding geometry, but making relation-level physical consistency calibrated, identity-preserving, and reportable. Remaining limits point to attachment-style relations, modern VLM sources, and downstream embodied-reasoning tests.

References

Chen, L.; Wang, X.; Lu, J.; Lin, S.; Wang, C.; and He, G. 2024. CLIP-Driven Open-Vocabulary 3D Scene Graph Generation via Cross-Modality Contrastive Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 27863–27873.

Feng, M.; Hou, H.; Zhang, L.; Wu, Z.; Guo, Y.; and Mian, A. 2023. 3D Spatial Multimodal Knowledge Accumulation for Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9182–9191.

Hou, H.-Y.; Lee, C.-Y.; Sonogashira, M.; and Kawanishi, Y. 2025. FROSS: Faster-Than-Real-Time Online 3D Semantic Scene Graph Generation from RGB-D Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 28818–28827.

Huang, I.; Bao, Y.; Truong, K.; Zhou, H.; Schmid, C.; Guibas, L.; and Fathi, A. 2025. FirePlace: Geometric Refinements of LLM Common Sense Reasoning for 3D Object Placement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13466–13476.

Koch, S.; Hermosilla, P.; Vaskevicius, N.; Colosi, M.; and Ropinski, T. 2024a. SGRec3D: Self-Supervised 3D Scene Graph Learning via Object-Level Scene Reconstruction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 3404–3414.

Koch, S.; Vaskevicius, N.; Colosi, M.; Hermosilla, P.; and Ropinski, T. 2024b. Open3DSG: Open-Vocabulary 3D Scene Graphs from Point Clouds with Queryable Objects and Open-Set Relationships. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14183–14193.

Liu, Y.; Mei, H.; Zhan, D.; Zhao, J.; Zhou, D.; Dong, B.; and Yang, X. 2026. View-on-Graph: Zero-Shot 3D Visual Grounding via Vision-Language Reasoning on Scene Graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7386–7394.

Madhavaram, V.; Sengar, V.; De, A.; and Sharma, C. 2026. VIZOR: Viewpoint-Invariant Zero-Shot Scene Graph Generation for 3D Scene Reasoning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 8584–8595.

Nguyen, M. A.; Tran, Q. H.; Le, B. N.; Pham, T. K.; and Guang, S. Y. 2026. RelWitness: Open-Vocabulary 3D Scene Graph Generation with Visual-Geometric Relation Witnesses. arXiv:2605.20823.

Sarkar, S. D.; Miksik, O.; Pollefeys, M.; Barath, D.; and Armeni, I. 2023. SGAligner: 3D Scene Alignment with Scene Graphs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 21927–21937.

Saxena, P.; and Chiun, J. 2025. ZING-3D: Zero-shot Incremental 3D Scene Graphs via Vision-Language Models. arXiv:2510.21069.

Shao, Y.; Zhai, W.; Yang, Y.; Luo, H.; Cao, Y.; and Zha, Z.-J. 2025. GREAT: Geometry-Intention Collaborative Inference for Open-Vocabulary 3D Object Affordance Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17326–17336.

Wald, J.; Dhamo, H.; Navab, N.; and Tombari, F. 2020. Learning 3D Semantic Scene Graphs From 3D Indoor Reconstructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3961–3970.

Wang, Z.; Cheng, B.; Zhao, L.; Xu, D.; Tang, Y.; and Sheng, L. 2023. VL-SAT: Visual-Linguistic Semantics Assisted Training for 3D Semantic Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 21560–21569.

Wang, Z.; Su, Y.; Li, C.; Wang, D.; Huang, Y.; Li, X.; and Zhao, B. 2025. Open-Vocabulary Octree-Graph for 3D Scene Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 7037–7047.

Xie, Y.; Pagani, A.; and Stricker, D. 2024. SG-PGM: Partial Graph Matching Network with Semantic Geometric Fusion for 3D Scene Graph Alignment and Its Downstream Tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 28401–28411.

Yu, F.; Deng, Q.; Tang, S.; Li, Y.; and Cheng, L. 2025. Open-World 3D Scene Graph Generation for Retrieval-Augmented Reasoning. Accepted by AAAI 2026, arXiv:2511.05894.

Zhang, C.; Delitzas, A.; Wang, F.; Zhang, R.; Ji, X.; Pollefeys, M.; and Engelmann, F. 2025. Open-Vocabulary Functional 3D Scene Graphs for Real-World Indoor Spaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 19401–19413.

Zhang, C.; Yu, J.; Song, Y.; and Cai, W. 2021. Exploiting Edge-Oriented Reasoning for 3D Point-Based Scene Graph Analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9705–9715.

Reproducibility Checklist

This checklist follows the AAAI-26 reproducibility checklist route checked on 2026-05-26. It is included after the references and is not part of the technical-content page count. Answers refer to the main paper and the Docker-based experiment artifact described in the paper.

General

- Conceptual outline and method description: yes. The paper describes the calibrated geometry-consistency framework, row contract, geometry join, operating points, and recall-violation protocol in Sections 3–4 and Figure 1.
- Separation between facts, claims, and speculation: yes. The paper states the scoped reliability claim, source-specific boundaries, blocked extensions, and limitations explicitly.
- Pedagogical and background references: yes. Related Work cites 3D scene graph prediction, open-vocabulary 3D scene graph methods, geometry-aware relation modeling, and reliability/calibration work.

Theoretical Contributions

- Does this paper make theoretical contributions? no.
- Formal assumptions, theorem statements, proofs, proof sketches, and theoretical-tool citations: NA.
- Empirical demonstrations of theoretical claims and code for theoretical counterexamples: NA.

Datasets

- Does this paper rely on one or more datasets? yes.
- Motivation for selected datasets: yes. Open3DSG is used as the main open-vocabulary relation-source case study, while VL-SAT is used as a controlled reproduced anchor under the same row contract and metrics.
- Novel datasets introduced by this paper: no.
- Novel dataset release and license: NA.
- Citations for existing datasets and sources: yes. The paper cites the relevant 3DSSG, VL-SAT, and Open3DSG sources.
- Public availability of existing datasets: partial. The method uses public research datasets and code routes, but large raw data, meshes, features, checkpoints, and model caches are subject to their original access terms and are not redistributed in the paper PDF.
- Non-public dataset description: NA; no new private dataset is introduced.

Computational Experiments

- Does this paper include computational experiments? yes.
- Development-time hyperparameter search and selection criteria: partial. The paper states the reported operating points, metric scope, and Open3DSG checkpoint selection by train-dev validation loss; exhaustive hyperparameter sweeps are not claimed.
- Preprocessing code: partial. The Docker experiment root records the preprocessing, feature, adapter, geometry-join, metric, table, and failure-analysis commands; large runtime artifacts are rebuilt or mounted rather than embedded in the PDF.
- Source code for experiments and analysis: partial. The tracked repository separates executable code under `src/`, Docker entry points under `configs/`, shell wrappers under `scripts/`, and compact outputs under `results/`; large datasets and model caches are intentionally external.
- Public code release: yes, planned upon publication, subject to anonymization during review and third-party dataset/model licenses.
- Implementation comments: partial. The experiment code and runbooks document the row contract, source adapters, geometry scoring, and metric generation, but final code-release cleanup is still required.
- Randomness and seeds: partial. The main evaluation pipeline is deterministic after fixed predictions, checkpoints, and manifests; full training randomness for reproduced source models is documented as run provenance rather than a multi-seed study.
- Computing infrastructure: partial. The Docker image, compose route, mounted paths, logs, and resource checks are recorded; final submission should include the exact GPU/CPU model and software versions used for the released artifact bundle.
- Evaluation metrics: yes. The paper defines exact-label R@K and Violation@K on the fixed $K = \{5, 10, 20, 50, 100\}$ grid, verifier GT/counterfactual metrics, controls, and failure-analysis criteria.
- Number of runs: yes. Reported source-result tables use one fixed reproduced source run per source/condition under locked manifests.
- Variation or distributional information beyond averages: partial. The paper reports source/family operating points, counterfactual verifier metrics, controls, structured audit, and failure rows, but it does not yet report repeated-run variance.
- Statistical significance tests: no. The current draft reports deterministic held-out recall and violation tradeoffs; no paired significance test is included.
- Final hyperparameters and operating points: partial. The paper and artifact reports list the final operating points, checkpoint choice, denominator policy, and source caveats. The paper-facing Open3DSG route is the 548/548 recovery-policy branch with `OPEN3DSG_MIN_VISIBLE_OBJECTS=2` and relaxed view regeneration for two scans; the exact-label measured-family denominator is 3,972, and residual calibration risk is reported through the failure-analysis artifacts. A final release table should collect every threshold and configuration in one place.